A.I Fundamentals

Module 1 Study Guide

Introduction to Artificial Intelligence

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: What AI Is — and What It Isn't

In this lesson: Define algorithm with local example; Define automation with local example; Def

Key points

- Algorithm = fixed procedure (interest formula)
- Automation = runs procedure automatically (nightly balance update)
- AI = learns & adapts (suggests savings product based on behavior)
- Understanding these helps you question tech claims

Examples from the lesson

- You are starting the course A.I Fundamentals, Module 1: Introduction to Artificial Intelligence, Lesson 1: What AI Is — and What It Isn't.
- For example, a credit union uses an algorithm to calculate monthly interest on a savings account: multiply the principal by the annual interest rate, then divide by twelve.
- If you deposit TT$5,000 at a 3 % annual rate, the algorithm works out TT$5,000 × 0.03 ÷ 12 = TT$12.50 interest for the first month.
- To see the three concepts together, let’s work through a concrete TT$ example.
- Imagine you open a fixed-deposit account at a local credit union for TT$10,000 with a 2 % annual interest rate, compounded monthly.
- The algorithm for monthly interest is: principal × annual rate ÷ 12.
- Plugging the numbers: TT$10,000 × 0.02 ÷ 12 = TT$16.67 interest earned each month.
- The credit union automates this calculation, so at the end of each month the system adds TT$16.67 to your balance without any staff involvement.
- After six months, your balance would be TT$10,000 + 6 × TT$16.67 = TT$10,100.02.
- In the next lesson, Lesson 2: A Brief History That Explains Today, we’ll explore how early ideas about machines thinking evolved into the AI tools we see in our markets, fetes, and credit unions today.

Lesson 2: A Brief History That Explains Today

In this lesson: Early Rule-Based Logic; Machine Learning Era; Generative AI Explosion; Local Fi

Key points

- Traced AI from rigid rule-based logic to modern generative tools
- Saw how big data and cheap computing hardware accelerated progress
- Measured how AI tools saved Kevin TT$ 1,600 monthly on business costs

Examples from the lesson

- I’m your instructor, and you are tuning in to our A.I Fundamentals course, Module 1: Introduction to Artificial Intelligence, for Lesson 2: "A Brief History That Explains Today".
- Imagine a local credit union using a system where a loan was automatically declined if your debt ratio was over forty percent, with zero flexibility for context.
- It was like a maxi taxi driver following one exact route—if there was sudden traffic on the Priority Bus Route, the system could not adapt or find a new way forward.
- Three main engines powered this boom: massive datasets from the internet, cheap and powerful graphics hardware, and breakthroughs in deep learning neural networks.
- In the early days of basic computing, Kevin spent TT$ 1,200 every month on software subscriptions, plus TT$ 800 hiring a data entry clerk to manually categorize every sales receipt.
- His total monthly expense for basic record-keeping was TT$ 2,000, and he had to wait until month-end to see his cash flow status.
- He now pays TT$ 150 a month for a unified cloud software tool, and TT$ 250 a month for an AI document assistant that automatically scans, categorizes, and organizes all his paper receipts.
- His new total software and technology expense drops to TT$ 400 per month.
- When we calculate the impact on Kevin's budget, his monthly operating expenses drop from TT$ 2,000 down to TT$ 400.
- That gives Kevin a net monthly cost savings of TT$ 1,600.
- It allows us to build smarter financial habits and manage daily tasks much faster than was ever possible ten years ago.
- We also saw how these shifts reduced technology costs for local business owners like Kevin from TT$ 2,000 down to TT$ 400, saving TT$ 1,600 every month.

Lesson 3: AI in Your Daily Life and Finances

In this lesson: AI in Everyday TT Banking; Fraud Shield vs Impulse Nudges; Monthly Budget Wo

Key points

- Smart algorithms protect transactions and deliver rapid support
- Frictionless commerce engines encourage unnecessary impulse purchases
- Conscious daily habit changes retain TT$600 monthly in savings

Examples from the lesson

- I’m glad you’re here for A.I Fundamentals, Module 1: Introduction to Artificial Intelligence, Lesson 3: AI in Your Daily Life and Finances.
- If your card normally handles a TT$150 grocery run in Tunapuna or a TT$40 doubles and lime stop, but suddenly registers a TT$3,000 online purchase from an unfamiliar foreign site at three in the morning, the system flags it instantly.
- Imagine your gross monthly income is TT$8,000 after standard taxes and NIS contributions.
- Fixed essential expenses—like your rent, utility bills, credit union savings, and weekly market groceries—take up TT$5,500 each month.
- That leaves you with a baseline disposable income of TT$2,500 for discretionary spending and flexible savings goals.
- Over a month, these small algorithmic nudges can add up to TT$1,200 in impulse purchases, while standard planned discretionary spending takes TT$800.
- Together, that discretionary payout totals TT$2,000, leaving you with just TT$500 in remaining cash surplus at the end of the month.
- You keep the fraud protection and fast customer service, but you turn off one-click purchasing, disable impulse shopping notifications, and set strict daily alert thresholds on your banking apps.
- By consciously curbing algorithmic impulse buys by half, you cut that impulse total down from TT$1,200 to TT$600.
- Your discretionary total drops to TT$1,400, which is TT$800 planned spending plus TT$600 controlled impulse spending.
- Subtracting TT$1,400 discretionary expenses from your TT$2,500 disposable income raises your remaining cash surplus from TT$500 up to TT$1,100 every single month.
- That extra TT$600 stays right in your local savings or emergency fund instead of slipping away through frictionless spending.
- Where AI helps is clear: fast fraud alerts, twenty-four-seven support, and quick transactions.
- Next time, we move into Module 2 to explore how you can actively prompt and partner with AI to build smarter budget systems.

Recap checklist — can you say each one?

Lesson 1: What AI Is — and What It Isn't

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- ☐ AI = learns & adapts (suggests savings product based on behavior)
- ☐ Understanding these helps you question tech claims

Lesson 2: A Brief History That Explains Today

- ☐ Traced AI from rigid rule-based logic to modern generative tools
- ☐ Saw how big data and cheap computing hardware accelerated progress
- ☐ Measured how AI tools saved Kevin TT$ 1,600 monthly on business costs

Lesson 3: AI in Your Daily Life and Finances

- ☐ Smart algorithms protect transactions and deliver rapid support
- ☐ Frictionless commerce engines encourage unnecessary impulse purchases
- ☐ Conscious daily habit changes retain TT$600 monthly in savings